Tuscany Pyroconvection – 3-Day Forecast – run 2026-07-20 00Z

Valid 2026-07-20 .. 2026-07-22. Hybrid: ICON-EU model levels + ICON-2I 2.2 km fuel gate. Method after Castellnou et al. (2022).

3-day forecast

Three valid days from the single 2026-07-20 00Z run (day 0, +1, +2), using the day+1/+2 forecast steps in the same ICON-EU + ICON-2I datasets. Atmosphere from ICON-EU native model levels (bulk-Richardson ABL, Bolton LCL, measured mixed-layer dtheta/dz, cap gamma-theta, ABL-top RH, shear); the 10 MW/m fuel gate uses ICON-2I's 2.2 km surface fields on the Tuscany .lcp fuels. See scripts/validation/README.md for the diagnostic validation (radiosondes + ERA5).

The **fuel-gated** map is the expected product; the **potential** map is the atmospheric upper bound (assumes a pyroCu-capable fire in every cell). Read class *counts* as indicative.

Fuel-gated (expected)

Potential (atmospheric upper bound)

Fuel-gated class distribution – daytime, % of land cells (13646 land cells)

Day	Hour	surface	convect	overshoot	resilient	deep
2026-07-20	09Z	100.0	0.0	0.0	0.0	0.0
2026-07-20	12Z	99.0	0.0	1.0	0.0	0.0
2026-07-20	15Z	97.9	0.2	1.9	0.1	0.0
2026-07-20	18Z	100.0	0.0	0.0	0.0	0.0
2026-07-21	09Z	99.7	0.0	0.3	0.0	0.0
2026-07-21	12Z	98.6	0.0	0.8	0.4	0.1
2026-07-21	15Z	97.2	0.0	1.5	0.4	0.9
2026-07-21	18Z	100.0	0.0	0.0	0.0	0.0
2026-07-22	09Z	99.1	0.0	0.8	0.1	0.0
2026-07-22	12Z	98.5	0.0	0.9	0.6	0.0
2026-07-22	15Z	97.9	0.0	1.0	1.1	0.0
2026-07-22	18Z	99.9	0.1	0.0	0.0	0.0

gated

Figure 1: gated

potential

Figure 2: potential

Classes: 0 surface plume, 1 convection plume, 2 overshooting pyroCu, 3 resilient pyroCu, 4 deep pyroCu/pyroCb. Generated by scripts/forecast_3day_tuscany.py.